HW05-AI Audit Log

Interaction 001 — Assignment intake and repository baseline

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 10:04:33 +07 (Asia/Ho_Chi_Minh)
Phase	PHASE 0 — Assignment Intake and Repository Baseline
Human prompt/instruction	Read the entire assignment PDF first; create assignment traceability and AI-audit scaffolding; verify Git remotes/status/branch and safe push default; inspect enough repository/SUT source to establish startup, dependency, and k6 status; do not select a workflow, create final k6 scripts, execute tests, commit, or push.
AI response summary	Read all 9 PDF pages; created a requirement traceability matrix and manual-responsibility register; verified repository remotes and branch; configured local `remote.pushDefault=origin`; inspected documentation, manifests, backend source/database, API clients, and installed-tool state; identified unresolved k6 artifact/video ambiguities and preliminary startup/dependency findings.
Human-review status	`WAITING FOR HUMAN` — phase findings and files require student review.
Corrections requested by human	None yet.

Files created

- `docs/assignment-requirements.md`
- `ai-audit/README.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md`

Files modified

- Repository-local Git configuration (`.git/config`): set

  `remote.pushDefault = origin`. No tracked SUT source file was modified.

Commands executed

Read-only inspection and verification commands included:

- `git remote -v`
- `git status --short --branch`
- `git branch --show-current`
- `git config --get remote.pushDefault`
- `git log -5 --oneline --decorate`
- `find`/`rg --files` repository structure queries
- `sed` reads of the README, setup guide, API specification, manifests,

`run_servers.sh`, `backend/server.js`, and `backend/database.js`

- `rg` route, authentication, environment, and frontend-client searches
- `file` and `ls -l` on the assignment PDF
- local Swift/PDFKit extraction of every PDF page (no package installed)
- `command -v` checks for PDF utilities, k6, and sqlite3
- `node --version` and `npm --version`
- `npm ls --depth=0` in all four JavaScript projects
- read-only SQLite table and row-count queries

Configuration command executed:

- `git config remote.pushDefault origin`

No server, Load, Stress, Spike, or endurance test was run. No database reset, dependency installation, commit, push, upload, issue publication, or screenshot creation occurred.

Evidence-backed observations

- `origin` is the student's expected repository and `upstream` is the official

  lecturer repository. Both currently display the expected HTTPS URL for fetch and push; policy forbids any push to `upstream`.

- Branch is `main`, tracking `origin/main`; the starting status contained only

  the untracked assignment PDF.

- The backend is Node.js/Express with SQLite, binds to port 3000, and is launched

  directly with `node server.js` according to source/setup documentation.

- No Docker configuration or `.env` file was found, and the backend source does

  not read `process.env`; the port and JWT signing secret are hard-coded. The latter is an existing SUT security concern and must not be copied into new performance-test data.

- `run_servers.sh` is not portable to this checkout: it contains another user's

  absolute path and begins with the destructive broad command `killall node`.

- Node.js `v20.20.2`, npm `10.8.2`, and `/usr/bin/sqlite3` are present. k6 is

  absent. No project has a `node_modules` directory; `npm ls --depth=0` reports all declared dependencies unmet.

- A populated SQLite database exists. It was inspected read-only and not reset.
- Source contradicts the stated FR-02 behavior: failed logins add 2 attempts and

  lock for 180 seconds, while the requirement says add 1 and lock for 30 seconds. This is a preliminary source finding, not an empirically reproduced issue.

- Credential documentation is inconsistent: the root README/database seed use

  the capitalized admin password, while `setup_guide.md` shows a lowercase form. The actual seeded database state must be verified safely before test design.

- Transactional state needs deeper investigation: carts live in process memory,

  checkout trusts the submitted total, creates persistent order rows, and does not visibly clear the cart in the backend route. Concurrency/repeatability and cleanup implications have not yet been designed.

Assumptions and limitations

- “Backend appears to run” is a source-based preliminary finding, not a verified

  startup result. Startup verification belongs to a later approved phase after dependencies are available.

- No workflow candidate has been evaluated or selected.
- The extracted PDF text was read for all pages. Formatting-aware page

  references use the PDF footer/page sequence reported by PDFKit.

- The `.jtl`/HTML requirement for k6 and the relationship between the two video

  descriptions remain unresolved; see `docs/assignment-requirements.md`.

Interaction 002 — Repository / SUT discovery

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 10:24:31 +07 (Asia/Ho_Chi_Minh)
Completed	2026-09-01 10:30:43 +07 (Asia/Ho_Chi_Minh)
Phase	PHASE 1 — Repository / SUT Discovery
Human prompt/instruction	Phase 0 was approved. Perform source-code-backed discovery only; do not install dependencies, start the SUT, mutate/reset/reseed the database, execute performance tests, select a workflow, or resolve the k6/video ambiguities. Fully document workflow-relevant endpoints, auth/JWT/lockout/recovery, products, cart, checkout/orders, coupons, seed state, side effects, correlation, concurrency, and CSV needs. Update the audit, create `docs/sut-discovery.md`, distinguish source facts from runtime verification needs, then stop for review. The exact prompt is preserved in `interactions/002-phase-1-repository-sut-discovery.md`.
AI output	Created a source-backed architecture, data-model, auth/lockout, route-contract, correlation, state-mutation, concurrency, CSV-data, ambiguity, and runtime-gap report covering all 31 backend routes. Found that server startup itself invokes the destructive database initializer, that UI carts do not call backend cart endpoints, and that checkout/order/coupon behavior has important state-isolation implications. All implementation discrepancies remain source observations rather than confirmed bugs.
Human-review status	`WAITING FOR HUMAN` — Phase 1 findings require student review before workflow candidates.
Corrections requested by human	None yet.

Phase 1 files created

- `docs/sut-discovery.md`
- `ai-audit/interactions/002-phase-1-repository-sut-discovery.md`

Phase 1 files modified

- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-001` set to `DONE BY HUMAN` from explicit approval;

  `H-026` added as the current review checkpoint)

Phase 1 commands and actions

- Read line-numbered backend server and database source with `nl`/`sed`.
- Read line-numbered web authentication, cart, product, checkout, profile,

  registration, password-reset, and routing source.

- Read relevant mobile and admin API-client source.
- Enumerated all Express route registrations with `rg`.
- Compared backend routes with API specification, setup guide, and requirements.
- Used `sqlite3 -readonly` for sanitized user state, product/category/coupon

  fixtures, order/coupon counts, schema/index inventory, PRAGMAs, and integrity.

- Calculated a SHA-256 hash and metadata for the existing SQLite file.
- Searched for environment configuration, JWT expiry/role checks, frontend API

  calls, and backend-cart client usage.

- Rechecked Git status and local time.
- One read-only `rg` client-call search failed because of unmatched shell

  quoting; a simpler successful search was used instead. It caused no mutation.

No dependency installation, server start, direct database initializer, write query, test execution, workflow selection, commit, push, or external action occurred.

Phase 1 verification

- An automated source/document comparison found 31 Express method/route

  registrations (24 distinct path strings) and zero undocumented paths.

- The SQLite SHA-256 remained

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`, with unchanged modification time `2026-09-01 09:51:44 +0700`.

- No `node_modules` directory appeared and `k6` remained unavailable.
- Whitespace and high-confidence private-key/token-pattern scans returned no

  finding in the Phase 1 documentation. The documented SUT demo credentials are public fixtures already present in the official repository, not private data.

Interaction 003 — Safe SUT startup strategy and runtime preparation

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 10:38:06 +07 (Asia/Ho_Chi_Minh)
Completed	2026-09-01 10:41:46 +07 (Asia/Ho_Chi_Minh)
Phase	PHASE 2 — Safe SUT Startup Strategy and Runtime Preparation
Human prompt/instruction	Phase 1 was approved. Reconfirm the reset-on-start control flow, affected database, and every-start behavior; do not modify the SUT, install anything, start the backend, mutate/reset the database, generate workflow candidates, run k6, commit, or push. Propose and compare 2–3 source-supported startup strategies, prefer behavior preservation, inspect dependencies/tool status, prepare exact macOS commands, create the startup runbook, preserve the actual interaction, separate source facts/proposals/runtime gaps, and stop at a human strategy checkpoint. The full prompt is preserved in `interactions/003-phase-2-safe-startup-strategy.md`.
AI output	Revalidated that `server.js` imports `database.js`, whose top-level `initDatabase()` call drops/recreates/seeds `backend/database.sqlite` on every fresh process. Compared original-checkout-with-backup, disposable local Git clone, and disposable backend-copy strategies. Recommended an exact-commit local clone under `/private/tmp`, which preserves application behavior while isolating reset/test mutations. Documented backend dependencies, installed prerequisites, absent k6/node_modules, affected paths, proposed install/start/read/stop commands, and runtime verification requirements. No SUT patch was proposed for execution.
Human-review status	`WAITING FOR HUMAN` — strategy and proposed commands require explicit approval.
Corrections requested by human	None yet.

Phase 2 files created

- `runbooks/sut-startup-macos.md`
- `ai-audit/interactions/003-phase-2-safe-startup-strategy.md`

Phase 2 files modified

- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-026` set to `DONE BY HUMAN` from explicit approval; `H-027` added as the current checkpoint)

Phase 2 inspections and commands

- Re-read line-numbered `backend/server.js`, `backend/database.js`, `setup_guide.md`, `run_servers.sh`, and backend manifests.
- Read direct dependency versions from `backend/package-lock.json` with a local Node one-liner; no module loading or installation occurred.

- Rechecked Node/npm/k6, Git, macOS utilities, Xcode/clang/make/Python, port

  3000 listener state, disk availability, and absence of `node_modules`.

- Inspected `sqlite3@6.0.1` engine/dependency metadata from the lockfile.
- Verified current Git commit, clean tracked/staged backend state, database Git

  blob identity, SHA-256, size, and modification time.

- Rechecked Git status, `remote.pushDefault`, and local timestamp.

No command from the runbook's `PROPOSED` execution blocks was run. No temp clone/backup, npm cache, `node_modules`, backend process, HTTP request, database write, source patch, k6 action, commit, or push was created/performed.

Phase 2 verification

- All seven proposed Bash command blocks passed `bash -n` syntax validation;

  this does not prove runtime success.

- Local Markdown-link validation found no broken file targets.
- The backend working tree remained clean and the original database hash/mtime

  remained unchanged.

- k6 and project `node_modules` remained absent, and port 3000 remained free.
- Whitespace and high-confidence secret-pattern scans found no issue in the

  created/updated Phase 2 documentation.

Interaction 004 — Approved safe startup verification

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 10:54:03 +07 (Asia/Ho_Chi_Minh)
Completed	2026-09-01 11:02:15 +07 (Asia/Ho_Chi_Minh)
Phase	PHASE 2B — Approved Safe Startup Verification
Human prompt/instruction	Strategy B approved: create a disposable `/private/tmp` Git clone pinned to the current commit; install npm dependencies only there; start only that clone; perform readiness and read-only product GETs; verify reset confinement and original integrity; stop the process; do not install k6, modify source, exercise mutations/auth failures, run performance tests, generate workflows, commit, or push. Preserve genuine failures and the exact interaction. The full prompt is in `interactions/004-phase-2b-runtime-startup-verification.md`.
AI output	Created a temp clone at the expected commit, installed locked dependencies after preserving an initial DNS failure, started the unmodified backend with PID 9739, proved it listened on port 3000 from the clone, received actual HTTP 200 responses from product search/listing, proved only the clone database reset, stopped the process, and verified the original database/source remained unchanged.
Human-review status	`WAITING FOR HUMAN` — Phase 2B results and unexpected environmental behavior require review.

Field	Record
Corrections requested by human	None yet.

Phase 2B files created

- `docs/runtime-startup-verification.md`
- `ai-audit/interactions/004-phase-2b-runtime-startup-verification.md`

Phase 2B files modified

- `runbooks/sut-startup-macos.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-027` set to `DONE BY HUMAN` from the explicit Strategy B

  approval; `H-028` added as the current review checkpoint)

Phase 2B commands and actions

- Recorded original repository identity, current commit, port state, backend

  working-tree state, file hashes, and absence of original `node_modules`.

- Created `/private/tmp/eshop-hw05-sut.jxn8Wd`, copied a verified original-DB

  backup into it, and made a `--no-hardlinks --local` clone at commit `85af3ba875c88283615e22cb108f13e2fccaf0e9`.

- Ran `npm ci` with a temp-local cache. The first sandboxed attempt failed with

  registry DNS `ENOTFOUND` and an incomplete dependency tree. Retrying the same command with approved network access succeeded; `npm ls --depth=0` and a native `sqlite3` load then succeeded.

- Started unmodified `node server.js`. Sandboxed launches were terminated by

  the execution environment; an approved localhost-listener execution remained active as PID 9739 from the clone and owned port 3000.

- Preserved sandbox curl HTTP 000/exit 7 results, then used approved localhost

  access for actual HTTP 200 product-search and product-list requests.

- Compared SQLite hashes/counts and confirmed startup removed the clone's prior

  three orders while leaving the original/backup byte-identical.

- Sent SIGTERM only to verified PID 9739, confirmed the listener disappeared,

  and ran read-only post-stop SQLite integrity/count checks.

- Rechecked original and clone Git state, backend hashes, original `node_modules` absence, and port state.

No system-wide package/Homebrew/sudo action, k6 installation/run, SUT patch, original database initialization, auth failure, registration, password reset, cart/coupon/order/admin mutation, performance execution, workflow generation, commit, or push occurred.

Phase 2B verification and unexpected results

- The original database SHA-256 remained

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6` before and after runtime execution; all original source/manifest hashes matched.

- The clone database hash changed to

  `e5783132d3952afb2bcc3410b33f2b05f0368f6542a6cb87f0e3fc98421ab992`, and its order count changed from 3 to 0, verifying clone-only reset/reseed.

- Both permitted API GETs returned HTTP 200 with genuine seeded-product JSON.
- npm reported four dependency vulnerabilities and one deprecated package; no

  automated remediation was performed.

- The first npm attempt and sandbox-local server/curl attempts failed because

  of environment restrictions. These observations are not classified as SUT bugs. The managed server session exited 1 after successful SIGTERM shutdown; the exact PID stopped, the port was free, and SQLite integrity was `ok`.

Interaction 005 — Controlled runtime API verification and workflow discovery

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 12:37:51 +07 (Asia/Ho_Chi_Minh)
Runtime completed	2026-09-01 12:42:32 +07
Documentation completed	2026-09-01 12:48:22 +07
Phase	PHASE 3 — Controlled Runtime API Verification and Workflow Discovery
Human prompt/instruction	Phase 2B accepted. Reuse only the disposable runtime; minimally and sequentially verify valid login/JWT/Bearer, product search/list/detail, optional category, cart append/read, checkout/order readback, and useful coupon behavior; preserve state/correlation/discrepancies and original integrity; then propose 5–6 genuinely distinct group workflows with full risk/ranking information. Do not test lockout, install k6, run performance tests, select a workflow, commit, or push. The complete prompt is preserved in `interactions/005-phase-3-runtime-api-workflow-discovery.md`.

Field	Record
AI output	Used PID 10568 from the approved clone for a sequential happy path. Observed HTTP 200 for login, Bearer profile, product/category reads, cart append/read, fixed coupon apply, checkout, usage recording, and order readbacks; correlated user 2, product 1, coupon 2, final amount 29,950,000, and order 1. Verified one clone order and usage row, persistent post-checkout cart content, and unchanged original repository. Proposed six distinct workflows and recommended WF-03 technically without selecting it.
Human-review status	`WAITING FOR HUMAN` — Phase 3 evidence/candidates, group uniqueness, and final workflow choice require explicit student decisions.
Corrections requested by human	None yet.

Phase 3 files created

- `docs/runtime-api-verification.md`
- `docs/workflow-candidates.md`
- `ai-audit/interactions/005-phase-3-runtime-api-workflow-discovery.md`

Phase 3 files modified

- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-028` set to `DONE BY HUMAN` from explicit acceptance;

  `H-029`, `H-004`, and `H-005` are active human checkpoints)

Phase 3 runtime actions and observations

- Reconfirmed original/temp commit identity, database hashes/counts, unlocked

  seed accounts, temp dependencies, clean original backend, and free port 3000.

- Restarted unmodified `node server.js` only in

  `/private/tmp/eshop-hw05-sut.jxn8Wd/repo/backend`; PID 10568 and its working directory/listener were verified with `lsof`.

- Sent a small sequential request set with `curl`; Node extracted JSON fields

  and redacted the transient JWT/password from stored evidence.

- Runtime-verified login and protected `/api/users/me`, product search/list/

  detail, categories, cart before/append/after, coupon list/application, checkout, usage insertion, history/detail, and post-checkout cart state.

- The fixed coupon response supplied `coupon_id=2` and

  `final_amount=29950000`; checkout supplied `orderId=1`. SQLite changed from zero to one order and zero to one coupon-usage row.

- Checkout retained the cart item and stored the submitted discounted amount.

Coupon apply and order detail returned 200 without Authorization. These are source/runtime observations, not automatically classified defects.

- Sent SIGTERM only to PID 10568. The session exited 143, PID disappeared, port

  3000 became free, and disposable SQLite integrity remained `ok`.

Phase 3 workflow output

Six business-process families were proposed for the four-person group:

1. WF-01 server-cart customer purchase; 2. WF-02 promotion redemption purchase; 3. WF-03 purchase and customer cancellation; 4. WF-04 new-customer onboarding and first order; 5. WF-05 administrative catalog import and verification; 6. WF-06 customer-to-admin fulfillment handoff.

WF-03 is the AI's best technical recommendation, conditional on a future single cancellation smoke verification. WF-01 is easiest; WF-06 offers the richest analysis. No workflow was selected.

Phase 3 safety/integrity

- Original database SHA-256 stayed

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`.

- Original backend status stayed clean, its source/manifest hashes remained

  unchanged, and original `node_modules` stayed absent.

- No failed login, lockout, registration, reset, cancellation, admin mutation,

  concurrent/high-volume request, SUT patch, system install, k6 action, performance run, commit, push, or external publication occurred.

Interaction 006 — Human workflow selection and functional smoke

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 12:56:28 +07 (Asia/Ho_Chi_Minh)
Runtime completed	2026-09-01 12:58:58 +07
Documentation completed	2026-09-01 13:00:58 +07
Phase	PHASE 4 — Selected Workflow Decision and Functional Smoke Verification

Field	Record
Human prompt /instruction	The student selected WF-03 and explicitly confirmed uniqueness among four group members. Record that as a HUMAN decision without reinterpretation; then use the approved disposable runtime for exactly one sequential login → product search/list → detail → checkout/create → cancel flow. Dynamically correlate JWT, product ID, and checkout order ID; preserve exact redacted evidence, before/after state, repeatability implications, and original integrity. Do not install/run k6 or perform any performance test. The complete prompt is preserved in [`interactions/006-selected-workflow-functional-smoke.md`](interactions/006-selected-workflow-functional-smoke.md).
AI output	Recorded WF-03 verbatim as the human choice. Reused the approved clone, reseeded only its temp database, and ran one functional iteration. Every request returned 200; search supplied product ID 1, checkout supplied order ID 1 dynamically, and correlated reads/cancellation proved `pending → canceled`. Stopped PID 11253, preserved one canceled temp order, and verified the original database/source remained unchanged.
Human-review status	`WAITING FOR HUMAN` — selected-workflow smoke evidence requires explicit review before detailed design or k6 work.
Corrections requested by human	None yet.

Phase 4 files created

- `docs/human-decisions.md`
- `docs/selected-workflow.md`
- `ai-audit/interactions/006-selected-workflow-functional-smoke.md`

Phase 4 files modified

- `docs/runtime-api-verification.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-004` and `H-005` set to `DONE BY HUMAN` from explicit selection/uniqueness statements; `H-030` added for smoke review)

Phase 4 runtime execution

- Preflight recorded original/temp commits, database hashes/counts, original backend status, free port 3000, temp dependencies, and absent k6.
- Starting unmodified `node server.js` in the temp clone reset its prior Phase 3 state to zero orders and zero coupon usages; PID 11253 owned port 3000.
- One seeded-customer login returned 200 and a 133-character JWT, which was redacted from repository evidence.
- Product search returned 200 and supplied product ID 1. Its dynamically built detail request returned 200 with matching ID and price 30,000,000.
- Checkout with the correlated JWT returned 200 and `orderId=1`. The actual value was extracted, not embedded as test data.

- A correlated pre-cancel read returned status `pending`; authenticated `PUT /api/orders/:orderId/cancel` returned 200; the correlated post-read returned `canceled`. The helper asserted that exact transition and exited 0.
- Read-only SQLite evidence confirmed one canceled order for user 2, unlocked accounts, zero coupon usage, and integrity `ok`.
- `kill -TERM 11253` stopped the verified clone PID. Its TTY session returned exit 1 after SIGTERM; the PID disappeared and port 3000 became free.

Phase 4 integrity and boundary

- Original database SHA-256 remained `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`.
- Original backend/source/manifest hashes remained unchanged, backend status stayed clean, and original `node_modules` stayed absent.
- The temp database intentionally retains one canceled order with hash `dea4a462e07ba1ec61450ceace2ef50ba2d9c9608d507129df14b0d5c75fb2fc`.
- This was functional smoke evidence only. No latency, throughput, VU, or other performance claim was made.
- No failed login, lockout, k6 action, Load/Stress/Spike/endurance run, source patch, system install, commit, push, or publication occurred.

Interaction 007 — Phase A selected workflow specification

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 13:05:55 +07 (Asia/Ho_Chi_Minh)
Documentation completed	2026-09-01 13:12:08 +07
Phase	PHASE A — Selected Workflow Specification
Human prompt/instruction	The student explicitly approved Phase 3 evidence and the WF-03 smoke, reaffirmed WF-03/its exact sequence/group uniqueness, and authorized H-029/H-030 as `DONE BY HUMAN`. Create a formal workload-independent contract for endpoints, group coverage, correlation/fail-fast behavior, iteration semantics, state/repeatability, lockout safety, data schemas, checks, future metric definitions, think-time locations, and shared-vs-scenario components. Do not install/run k6, create final scripts, choose workload levels, or perform testing. The full prompt is preserved in `interactions/007-phase-a-selected-workflow-specification.md`.

Field	Record
AI output	Recorded both approvals as human decisions and closed H-029/H-030. Created the WF-03 formal contract: five immutable business steps plus correlated pending/canceled probes; dynamic JWT/product/price/order chain; iteration-local fail-fast rules with no static fallback; verified state implications; secret-safe proposed CSV split; meaningful business assertions; future native/custom metrics all marked not measured; candidate think-time locations without values; and workload-independent architecture.
Human-review status	`WAITING FOR HUMAN` — H-031 requires explicit Phase A review before Phase B.
Corrections requested by human	None yet.

Phase A files created

- `docs/selected-workflow-specification.md`
- `ai-audit/interactions/007-phase-a-selected-workflow-specification.md`

Phase A files modified

- `docs/selected-workflow.md`
- `docs/human-decisions.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-029` and `H-030` set to `DONE BY HUMAN` exactly as

  authorized; `H-031` added for Phase A review)

Phase A contract decisions

- Kept the human-selected sequence unchanged. Two `GET /api/orders/:orderId`

  probes validate pending/canceled state around the five business steps.

- Classified login as auth-heavy, product search/detail as read-heavy, and

  checkout/cancellation as transactional.

- Required iteration-local JWT, product ID, validated price, and new order ID.

  Missing correlation aborts the iteration; seed/static/prior-iteration fallbacks are forbidden.

- Defined success as all step and lifecycle checks passing for the same order.

  A checkout followed by failed cancellation can leave a pending row and must remain a visible workflow failure.

- Recorded one persistent canceled order per successful iteration, no inventory

  behavior, database growth, potential unmeasured SQLite contention, and an evidence-first disposable-reseed strategy for comparable later scenarios.

- Required valid credentials only, no guessing/retry/lockout experiment, clear unexpected 401/403 metrics, and no token/secret leakage.

- Proposed a committed non-secret workflow CSV plus ignored/private credential

  mapping. Multiple valid customer accounts are recommended but not yet available or approved.

- Defined native/custom metric meanings with every future value explicitly

  `NOT MEASURED`; no numeric threshold was selected.

- Identified four candidate think-time points without assigning durations.
- Required one shared workflow implementation for Load/Stress/Spike; only

  later workload/output configuration may differ.

Phase A boundary and verification

No SUT process or HTTP request was executed. No data file, k6 script, workload, think-time value, threshold, result, dependency installation, commit, push, or publication was created/performed. The work was documentation-only and based on human-approved source/runtime evidence.

Subsequent Phase A human review

- Received: 2026-09-01 13:26:24 +07.
- Outcome: Phase A approved with one required clarification; H-031 explicitly

  authorized as `DONE BY HUMAN`.

- Human correction: the two order-detail HTTP calls are invariant executable

  validation substeps. They affect request count/latency/iteration duration and must not be described as having no effect on the executable workflow.

- Corrected invariant sequence: login → product search/list → product detail →

  checkout → verify pending → cancel exact order → verify canceled.

- Load, Stress, and Spike must all execute that same seven-request sequence;

  only workload models may differ.

- The correction is recorded in `docs/human-decisions.md` as HD-003 and appended

  to the detailed Interaction 007 record without erasing the original output.

Interaction 008 — Phase B workload model proposal

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 13:26:24 +07 (Asia/Ho_Chi_Minh)

Field	Record
Documentation completed	2026-09-01 13:31:51 +07
Phase	PHASE B — Workload Model Proposal
Human prompt/instruction	Apply the Phase A human correction, close H-031, and propose concrete but unvalidated Load/Stress/Spike models using the identical seven-request WF-03 sequence. Include think times, bounded executors/stages/recovery, model-only iteration/order estimates, scenario isolation, account analysis, safety-vs-threshold distinction, executor comparison, and a blank human-review table. Do not install/run k6, start the SUT, generate final scripts, finalize thresholds, or call assumptions measured. The complete prompt is preserved in `interactions/008-phase-b-workload-model-proposal.md`.
AI output	Created an assumption-labelled proposal: 5-VU sustained Load plus a 2-VU alternative; progressive Stress capped at 20 VUs; 3→20-VU rapid Spike with recovery; 3–6 seconds total intentional think time and immediate lifecycle cancellation; VU-second-based state estimates explicitly not measured; fresh disposable scenario isolation; one-account initial feasibility with multiple accounts deferred to humans; operational safety guardrails separated from unset performance thresholds; and executor rationale. All review decisions/final values remain blank.
Human-review status	`WAITING FOR HUMAN` — H-032 and the workload review table require explicit corrections/approval before Phase C.
Corrections requested by human	None yet for Phase B.

Phase B files created

- `docs/workload-model-proposal.md`
- `ai-audit/interactions/008-phase-b-workload-model-proposal.md`

Phase B files modified

- `docs/selected-workflow-specification.md`
- `docs/selected-workflow.md`
- `docs/human-decisions.md`
- `ai-audit/interactions/007-phase-a-selected-workflow-specification.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md` (`H-031` set to `DONE BY HUMAN`; `H-032` added for workload review)

Phase B AI proposals

- Shared think time: uniform 0.5–1.0 seconds after login, 1.0–2.0 after search, and 1.5–3.0 after detail; immediate cancellation after pending verification.
- Load primary: `ramping-vus`, 0→5 over one minute, five VUs for five minutes, 5→0 over one minute; conservative two-VU alternative.
- Stress primary: `ramping-vus`, progressive 2/5/10/15/20-VU ramps/holds,

bounded at 20, then five-VU recovery and ramp-down.

- Spike primary: `ramping-vus`, three-VU baseline, 3→20 in ten seconds,

  45-second spike, 20→3 in ten seconds, two-minute recovery.

- All use 30-second graceful ramp-down/stop proposals and the same seven-request

  workflow.

- Integrated VU-seconds were calculated mechanically: Load 1,800, conservative

  Load 420, Stress 6,810, Spike 1,940. Estimated iteration/order ranges use an explicitly unmeasured 4–10-second iteration assumption and are not throughput.

- Proposed a fresh commit-pinned disposable runtime and unique artifact path for

  each scenario, preserving evidence before any reseed.

- One valid customer is initially feasible but shares login/user-row state;

  provisioning multiple accounts remains a future human decision.

- Operational aborts/caps were labelled safety guardrails. No p90/p95/p99,

  error-rate, throughput, capacity, or regression performance threshold was set.

Phase B boundary

No SUT/backend process, HTTP request, k6 installation/execution, final script, data file, measured value, final workload, performance threshold, commit, push, or publication was created/performed.

Subsequent Phase B human review

- Received/recorded: 2026-09-01 13:37:42 +07.
- The human's session-end instruction explicitly identifies Phase B as reviewed

  and requests the Load/Stress/Spike/think-time/account/isolation/safety values as human-approved decisions in the resume checkpoint.

- H-032 was therefore set to `DONE BY HUMAN`; the exact approved planning values

  are recorded in `docs/CODEX-RESUME-CHECKPOINT.md` and HD-004.

- The values remain unmeasured planning inputs, not production traffic,

  capacity, actual throughput/state growth, or final performance thresholds.

- Phase C was explicitly not started.

Interaction 009 — Session-end resume checkpoint

Field	Record
AI tool	Codex CLI

Field	Record
Date/time	2026-09-01 13:37:42 +07 (Asia/Ho_Chi_Minh)
Documentation completed	2026-09-01 13:39:27 +07
Phase context	Phase B complete; Phase C not started
Human prompt/instruction	Create `docs/CODEX-RESUME-CHECKPOINT.md` for the next session with human-reviewed Phase B status, WF-03 and its exact sequence, approved workload/think-time/account/isolation/safety values, unresolved items, authoritative files, completed decisions, next intended Phase C, and critical safety/audit rules. Do not start Phase C; update the AI Audit and stop. The complete prompt is preserved in `interactions/009-session-end-resume-checkpoint.md`.
AI output	Created an authoritative resume checkpoint at commit `85af3ba8...`, recorded exact Phase B planning decisions while retaining their unmeasured status, listed unresolved work and continuation files, preserved Git/database/disposable-runtime/k6/human-checkpoint rules, recorded HD-004/H-032, and explicitly left Phase C not started.
Human-review status	Checkpoint explicitly requested by human; Phase B recorded reviewed. Future Phase C requires a new explicit instruction.
Corrections requested by human	None for the checkpoint at creation time.

Session-end files created

- `docs/CODEX-RESUME-CHECKPOINT.md`
- `ai-audit/interactions/009-session-end-resume-checkpoint.md`

Session-end files modified

- `docs/human-decisions.md`
- `MANUAL-TODO.md` (`H-032` set to `DONE BY HUMAN`)
- `ai-audit/audit.md`

Session-end verified state

- Commit remained `85af3ba875c88283615e22cb108f13e2fccaf0e9`.
- `origin` and `upstream` retained expected URLs; `remote.pushDefault=origin`; pushing upstream remains forbidden.
- Port 3000 was free and k6 remained uninstalled.
- Original database SHA-256 remained `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`.
- No runtime, database mutation, Phase C artifact, dependency installation, commit, or push occurred.

Interaction 010 — Resume context recovery and Phase B discrepancy

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 15:51:08 +07 (Asia/Ho_Chi_Minh)
Phase context	Resume reconstruction before Phase C
Human prompt/instruction	Reconstruct the project only from ordered repository evidence; verify identity, workflow, completed phases, and approved Phase B values; audit the new session; proceed to Phase C only if recovery succeeds; stop and show any repository/resume-summary difference. The complete prompt is preserved in `interactions/010-resume-context-recovery-discrepancy.md`.
AI output	Read all nine assignment-PDF pages and every requested context source in order, plus recent Interactions 007–009 and required Git/OS/tool checks. Most checkpoint facts matched, but the prompt conflicted with the human-approved repository on pending-to-cancel think time and the approved account strategy. Applied the explicit stop rule and did not begin Phase C.
Human-review status	**RESOLVED BY HUMAN REVIEW** on 2026-09-01 15:58:08 +07; both conflicting values were explicitly declared new corrections.
Corrections requested by human	Preserve the old values, then supersede them with pending→cancel random 0.5–1.0 s and a design for up to 20 disposable-runtime dedicated accounts.

Resume verification result

- Confirmed student Phạm Ngọc Gia Bảo (`23127027`), tool k6, macOS, WF-03,

  its seven-request executable sequence, four-person-group uniqueness, and the recorded completion/review status through Phase B. Phase C remains not started.

- Recovered the repository's exact Load, Stress, and Spike planning timelines.

  They remain unmeasured planning inputs, not capacity or final thresholds.

- Confirmed fresh comparable disposable runtime isolation per official

  scenario and evidence preservation before reset.

- Git remained on `main` at `85af3ba`, tracking `origin/main`, with expected

  remotes and `remote.pushDefault=origin`. k6 remained absent.

Discrepancy and boundary

1. The prompt states random `0.5–1.0 s` from pending verification to cancellation; the authoritative checkpoint records `0 s` immediate lifecycle completion. 2. The prompt states a dedicated-account strategy targeting up to 20 accounts; the authoritative checkpoint approves one seeded customer initially and leaves multi-account provisioning as a future human decision.

Because the prompt explicitly required stopping on any difference, no `docs/test-data-strategy.md`, Phase C design, private credential file, provisioning, `.gitignore` rule, or `MANUAL-TODO.md` change was made. No SUT, database, k6, performance-test, commit, push, or external action occurred.

Subsequent resolution

The human explicitly resolved both discrepancies as new Phase B corrections. The historical values remain in Interaction 010 and earlier records; HD-005 and Interaction 011 contain the superseding decision. This later resolution does not rewrite the fact that Codex correctly stopped before receiving it.

Interaction 011 — Phase B corrections and Phase C test-data strategy

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 15:58:08 +07 (Asia/Ho_Chi_Minh)
Documentation completed	2026-09-01 16:06:13 +07
Phase	Phase B correction resolution; PHASE C — Test-Data Strategy
Human prompt/instruction	Declare the two differences intentional new human corrections, preserve prior values, update authoritative Phase B records, resolve the discrepancy checkpoint, and design Phase C public/private data, disposable provisioning for up to 20 customers, deterministic VU mapping, product/address/quantity inputs, correlation boundaries, secret handling, and comparable preconditioning. Do not provision accounts, create real credentials, install k6, generate final scripts, or execute tests. The complete prompt is preserved in `interactions/011-phase-b-corrections-phase-c-test-data-strategy.md`.
AI output	Preserved the old `0 s` and one-seeded-account values, recorded HD-005, and made random 0.5–1.0 s plus an up-to-20 dedicated disposable-account design current. Source-backed Phase C uses sequential registration outside measurement, deterministic VU/account rows, response-owned dynamic correlation, runtime-only credentials, strict pool validation, safe product/address/quantity data, narrow proposed ignore rules, and identical fresh preconditioning for all scenarios.
Human-review status	**APPROVED WITH CORRECTIONS** on 2026-09-01; H-033 set `DONE BY HUMAN` only after correction/artifact validation.
Corrections requested by human	Verify quantity against real WF-03 and remove it if unused; create approved safe artifacts; reinforce provisioning as separate preconditioning before measured k6.

Source evidence used

- `backend/server.js`: registration, login, product search/detail, checkout, cancellation, and order-detail behavior.
- `backend/database.js`: reset-on-start, users schema/default role, seed users, products, orders, and absence of email uniqueness.
- Read-only original SQLite schema/seed inspection; no mutation.
- Existing Phase A/B workflow, correlation, isolation, human decision, resume, and audit records.
- Current repository ignore state: `.gitignore` is absent.

Files created

- `docs/test-data-strategy.md`
- `ai-audit/interactions/011-phase-b-corrections-phase-c-test-data-strategy.md`

Files modified

- `docs/workload-model-proposal.md`
- `docs/CODEX-RESUME-CHECKPOINT.md`
- `docs/human-decisions.md`
- `docs/selected-workflow-specification.md` (Phase C supersession note; original

  Phase A data proposal retained)

- `ai-audit/interactions/010-resume-context-recovery-discrepancy.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md`

Phase C outcome and boundary

- Planned public schema:

  `row_id,account_key,search_term,expected_product_name,quantity,shipping_address`.

- Private runtime schema:

  `account_key,email,password,expected_role`.

- Account keys map directly from VU 1..20 to slots 01..20; Load activates the

  first five, Stress/Spike can activate all 20, and the pilot uses the first two.

- Public rows use exact reviewed seed-product matching, quantity one, and

  synthetic per-slot addresses.

- JWT, user ID, product ID, price, checkout total, and order ID are runtime

  values; response-owned values are forbidden in CSV.

- Official scenarios fail preflight if the complete 20-account pool is not

  validated. No shared/seeded fallback or silent peak reduction is allowed.

- Registration and account-validation traffic are setup and excluded from

  measured WF-03 artifacts.

No account, credential file, password, JWT, public CSV, `.gitignore` change, provisioning helper, SUT process, HTTP request, database mutation, k6 action, final scenario script, performance result, commit, push, or publication was created/performed.

Subsequent Phase C human review

- The human approved the overall design and explicitly authorized H-033 after corrections were applied.

- Quantity was verified against source and the selected smoke: checkout accepts

  only amount/address, the smoke used detail price directly, and the flow has no quantity consumer. It was removed from the final public schema.

- The human approved 20 public rows, private schema/template, separate

  preconditioning, complete pool validation, dedicated VU mapping, dynamic correlation, narrow ignore rules, and equivalent scenario setup.

- The original AI proposal, human review, and final decision are preserved in

  `docs/test-data-strategy.md`, HD-006, Interaction 011, and Interaction 012.

Interaction 012 — Phase C review and Phase D correlation strategy

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 16:15:12 +07 (Asia/Ho_Chi_Minh)
Documentation completed	2026-09-01 16:19:22 +07
Phase	Phase C correction/finalization; PHASE D — Authentication and Dynamic Correlation Strategy
Human prompt/instruction	Approve Phase C with a mandatory quantity-use validation, safe creation of the 20-row public CSV/template/README/narrow ignore rules, strict separation of account setup from official metrics, complete pool validation and dedicated VU mapping; mark H-033 done only after corrections; then design Phase D dynamic correlation, ownership/lifetime/failure behavior, VU isolation, and stable request grouping/tagging without implementation or execution. The complete prompt is preserved in [`interactions/012-phase-c-review-phase-d-correlation-strategy.md`](interactions/012-phase-c-review-phase-d-correlation-strategy.md).
AI output	Proved quantity unused and removed it; created/validated the authorized safe artifacts; recorded HD-006 and H-033 complete; verified official k6 execution-ID documentation; created a seven-step correlation/fail-fast/isolation/tagging strategy and opened H-034 for human review.
Human-review status	Phase C `DONE BY HUMAN`; Phase D `WAITING FOR HUMAN`.
Corrections requested by human	None yet for Phase D.

Evidence and verification

- Source checkout reads only `total_amount` and `shipping_address`; selected

  WF-03 smoke submitted only those fields and used normalized detail price.

- `workflow.csv` passed exact header, 20-row, sequential key, five-product

  rotation, synthetic address, and uniqueness validation.

- `credentials.template.csv` passed exact private header and placeholder-only

  validation. No local/private credential file exists.
- All four approved private paths are ignored; each public CSV/template/README

  was checked individually and remains trackable.
- The first multi-path `git check-ignore --quiet` validation emitted the genuine

  error that quiet mode accepts one pathname. The corrected per-file checks all passed; the failed command is preserved rather than concealed.
- Official Grafana k6 documentation for `k6/execution` and data parameterization

  was read to verify `exec.vu.idInTest` as globally unique/test-wide and suitable for per-VU data binding. No tool installation occurred.

Files created

- `.gitignore`
- `performance/data/workflow.csv`
- `performance/data/credentials.template.csv`
- `performance/data/README.md`
- `docs/correlation-strategy.md`
- `ai-audit/interactions/012-phase-c-review-phase-d-correlation-strategy.md`

Files modified

- `docs/test-data-strategy.md`
- `docs/selected-workflow-specification.md`
- `docs/CODEX-RESUME-CHECKPOINT.md`
- `docs/human-decisions.md`
- `ai-audit/audit.md`
- `MANUAL-TODO.md`

Phase D outcome and boundary

- Proposed VU source: `exec.vu.idInTest`, validated in `1..20`, with direct

  public/private row binding and no fallback.
- Per-step design covers login/JWT, search/product ID, detail/price, checkout/

  order ID, pending check, corrected cancellation pause, same-order cancel, and final canceled check.
- Every dynamic value has explicit source, validation, lifetime, owner, and

  failure behavior; mutable correlation state is iteration-local only.
- Stable groups/tags distinguish all seven measured requests without dynamic

  IDs, secrets, or added API calls.
- Proposed next phase: Phase E — Checks, Custom Metrics, and Safety-Stop

Specification, subject to H-034 review.

No account, real credential/password/JWT, provisioning helper/run, SUT process, HTTP request, database mutation, k6 action, final scenario script, performance result, commit, push, or publication was created/performed.

Subsequent Phase D human review

- The human approved the mapping guard, exact product match, post-checkout

  evidence policy, groups/tags, and response-correlation lifetimes.
- The human corrected the original AI proposal that the first unexpected login

  `401` / `403` should stop the whole test. The current decision fails only that isolated iteration; a test-level abort is reserved for confirmed lockout or clear systemic/runtime danger.
- The original proposal, human review, and final decision remain visible in

  `docs/correlation-strategy.md`, HD-007, Interaction 012, and Interaction 013.
- H-034 was set to `DONE BY HUMAN` only after these records were applied.

Interaction 013 — Phase D review and Phase E checks, metrics, safety, and reports

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 16:29:28 +07 (Asia/Ho_Chi_Minh)
Documentation completed	2026-09-01 16:33:23 +07
Phase	Phase D correction/finalization; PHASE E — Checks, Custom Metrics, Safety Stops, and Report/Output Proposal
Human prompt/instruction	Approve Phase D with guarded separate-run `exec.vu.idInTest` mapping, exactly one exact product match, visible post-checkout residual lifecycle evidence, stable tags/lifetimes, and a correction making an isolated unexpected login `401` / `403` an iteration failure rather than an automatic whole-test abort; then design Phase E exact checks, low-cardinality metrics, one outcome per iteration, failure taxonomy, review-required operational safety stops, and an explicitly unresolved three-output/JTL/HTML proposal. The complete actual prompt is preserved in `interactions/013-phase-d-review-phase-e-checks-metrics-reports.md`.
AI output	Preserved the old auth proposal, human correction, and final decision; recorded HD-007 and H-034 completion; created exact seven-step checks, a minimal custom-metric/one-outcome design, failure taxonomy and operational guardrail proposal; documented both assignment output clauses, current official k6 output capabilities, three distinct primary outputs, authentic raw-data rules, two unresolved compliance paths, and an exact lecturer/TA question.
Human-review status	Phase D `DONE BY HUMAN`; Phase E H-035 `WAITING FOR HUMAN`; k6 `.jtl`/HTML H-002 `UNRESOLVED`.

Field	Record
Corrections requested by human	Replace “abort whole test on first unexpected login 401/403” with isolated iteration failure; retain test-level safety abort only for confirmed lockout/systemic unsafe execution.

Evidence and documentation research

- Read the existing authoritative correlation, human-decision, workload,

  resume, assignment-requirement, manual, and AI-audit records.
- The local attempt to use `pdftotext` failed because that command is absent;

  no package was installed. Existing PDF traceability in `docs/assignment-requirements.md` was used to reference T1-07 and T1-14.
- Current official Grafana k6 documentation was checked for native granular

  JSON/CSV, multiple outputs, end/custom summaries, and web-dashboard HTML export. No k6 binary was installed or executed.
- The official supported-output list does not document native JMeter JTL; this

  is recorded as a documentation-based inference that must be rechecked against the future pinned version, not as fabricated execution evidence.
- Final read-only validation found branch `main` at `85af3ba`, tracking

  `origin/main`; remotes `origin` and `upstream` unchanged; and `remote.pushDefault=origin`.
- Original `backend/database.sqlite` remained SHA-256

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`; port 3000 had no listener; k6 remained absent.
- Public workflow data retained its approved header and 20 rows; the credential

  template retained only its header and one placeholder example. No local credential file exists. The four private paths are ignored and public artifacts remain trackable.

Files created

- `docs/checks-metrics-safety.md`
- `docs/report-output-mapping.md`
- `ai-audit/interactions/013-phase-d-review-phase-e-checks-metrics-reports.md`

Files modified

- `docs/correlation-strategy.md`
- `docs/workload-model-proposal.md`
- `docs/CODEX-RESUME-CHECKPOINT.md`
- `docs/human-decisions.md`
- `ai-audit/audit.md`

- `MANUAL-TODO.md`

Phase E proposed outcome and boundary

- Native tagged `http_req_duration` provides the seven endpoint Trends; custom

  duplicates are deliberately avoided.
- A custom attempted Counter plus exactly one final Boolean workflow Rate sample

  provides the business denominator; a single tagged failure Counter records at most the first terminal workflow failure.
- Created/canceled Counters expose incomplete lifecycle residue; cancellation

  counts only after the final same-order canceled probe.
- Six failure classes distinguish transport/protocol, authentication,

  correlation/data, business assertion, lifecycle, and runtime/safety effects.
- Every numeric operational abort candidate is explicitly marked for human

  review and is not a performance threshold. A first isolated auth response is explicitly excluded from test-level abort behavior.
- Candidate primary outputs are Load end summary, Stress native CSV time

  series, and Spike native web-dashboard HTML, with native granular JSON as the canonical raw artifact for each scenario.
- Literal `.jtl` and three-HTML-folder compliance remains unresolved; no data

  may be renamed or converted and presented as native JMeter JTL.

No account, real credential/password/JWT, provisioning helper/run, SUT process, HTTP request, database mutation, k6 action, final scenario script, performance result, JTL/HTML result, report dependency, commit, push, or publication was created/performed.

Subsequent Phase E human review

- The human approved the exact checks, minimal custom/native metric model,

  guarded one-outcome semantics, six-class taxonomy, immediate unsafe-runtime stops, 2 GiB disk minimum, and bounded workload/wall-clock/graceful controls.
- The human explicitly deferred four numeric AI abort proposals—5 connection

  failures, 5 cross-account auth failures, 10 transactional 5xx responses, and 20% failures across two 15-second windows—until controlled pilot/runtime evidence. They remain history and are not implemented.
- The Load-summary/Stress-CSV/Spike-dashboard mapping was conditionally approved

  as a technical proposal only. H-002 remains waiting for lecturer/TA; the question has not been sent and no response exists.

- H-035 was set to `DONE BY HUMAN` only after HD-008 and authoritative records were updated.

Interaction 014 — Phase E review and Phase F static k6 architecture

Field	Record
AI tool	Codex CLI
Date/time	2026-09-01 16:41:04 +07 (Asia/Ho_Chi_Minh)
Documentation/static review completed	2026-09-01 16:54:05 +07
Phase	Phase E correction/finalization; PHASE F — Shared k6 Architecture and Static Implementation Review
Human prompt/instruction	Approve Phase E checks/metrics/outcome/taxonomy/immediate safety and bounded disk/time/VUs; defer four numeric error-abort rules pending pilot evidence; conditionally approve the output mapping while H-002 waits unsent for lecturer/TA; then create a clearly draft, shared, static-only k6 architecture with one WF-03 implementation and exact workloads, without installing/running k6 or the SUT, provisioning accounts, or creating results. The complete actual prompt is preserved in `interactions/014-phase-e-review-phase-f-static-k6.md`.
AI output	Recorded HD-008/H-035 and H-002 waiting state; created the TA register; created a compact config/data/check/metric/workflow/scenario/output architecture with one business implementation and three thin entry points; represented exact Load/12m30s Stress/Spike plans; statically checked grammar, shared-flow structure, tags, guards, secrets, thresholds, and correlation; opened H-036.
Human-review status	Phase E `DONE BY HUMAN`; Phase F H-036 `WAITING FOR HUMAN`; H-002 `WAITING FOR LECTURER/TA CLARIFICATION`.
Runtime verification	**NONE — draft source is not k6/runtime verified**

Files created

- `performance/config/workloads.js`
- `performance/config/runtime.js`
- `performance/config/output.js`
- `performance/lib/csv.js`
- `performance/lib/data.js`
- `performance/lib/auth.js`
- `performance/lib/checks.js`
- `performance/lib/metrics.js`
- `performance/lib/workflow.js`
- `performance/scenarios/load.js`
- `performance/scenarios/stress.js`
- `performance/scenarios/spike.js`
- `performance/tools/README.md`
- `docs/k6-architecture.md`
- `docs/ta-clarifications.md`
- `ai-audit/interactions/014-phase-e-review-phase-f-static-k6.md`

Files modified

- `docs/checks-metrics-safety.md`
- `docs/report-output-mapping.md`
- `docs/correlation-strategy.md`
- `docs/workload-model-proposal.md`
- `docs/CODEX-RESUME-CHECKPOINT.md`
- `docs/human-decisions.md`
- `performance/data/README.md`
- `MANUAL-TODO.md`
- `ai-audit/audit.md`

Static findings and genuine validation output

- Two initial combined validation commands failed before their checks with the

  genuine zsh message `unmatched "` due command quoting. The commands were corrected; the failure is preserved here rather than hidden.

- General JavaScript grammar passed `node --check` for all 12 draft `.js` files.

  This is not a k6 compatibility or execution claim.

- Static extraction counted 38 unique approved stable check names and found all

  seven approved custom metric names.

- Normalizing only scenario names made the three thin entry files produce the

  same SHA-256; no `/api/` endpoint occurs in them. All seven endpoint references are in shared `performance/lib/` code, and all business sequencing remains in the one shared `workflow.js` orchestrator.

- No final threshold/deferred numeric abort code, static JWT/product ID/order ID,

  scenario-specific business flow, modulo/account fallback, or real credential artifact was found.

- The Stress array contains 13 stages totaling 750 seconds, matching 12m30s.
- Stable request names/steps are used and native `url` system tags are excluded

  to prevent correlated order paths becoming metric labels.

- Expected private paths remain ignored. Public data/templates/docs remain

  trackable.

- Original `backend/database.sqlite` SHA-256 remained

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6`; k6 remained absent; port 3000 had no listener.

Phase F boundary

- The source uses current documented k6 interfaces but is explicitly **DRAFT —

NOT RUNTIME VERIFIED**. `node --check` cannot validate k6 imports, executor behavior, `open()` path semantics, `SharedArray`, `exec.test.abort`, HTTP behavior, output overhead, or SUT contracts.

- No final p95/RPS/error/capacity thresholds or four deferred numeric abort

  conditions are coded.

- External commit/database/account/disk/PID/port/evidence/wall-clock preflight

  and runner logic is intentionally not claimed or implemented yet.

- Output mapping is metadata isolated from the workflow; no JTL conversion or

  compliance claim exists.

No k6 install/run, SUT process, HTTP request, account, credential, provisioning, database mutation, performance result, raw output, report, screenshot, commit, push, or publication was created/performed.

Subsequent Phase F human review and Git recovery authorization

- The human approved the one shared workflow, workload-only scenario entries,

  static checks/metrics/outcome/failure implementation, dedicated account mapping, and immediate safety model without claiming runtime compatibility.

- The four numeric error-abort proposals and all final performance thresholds

  remain deferred/undefined.

- The human resolved H-002 by selecting genuine distinct k6-equivalent outputs

  under the PDF. The earlier AI concern and unsent question remain preserved; no lecturer/TA response is invented and no JMeter artifact is fabricated.

- Git inspection showed all Phase 0–F artifacts had been developed before local

  procedural commits. The human authorized truthful logical baseline commits, prohibited fake/backdated history, and prohibited push.

- Local baseline commits created so far:

  `e488cf4` discovery/runtime evidence; `e0d2df5` workflow/workload/data; `3f1e869` correlation/checks/reporting; `1674905` shared draft k6 source.

- Each commit's staged names/stat/check, forbidden path check, and secret scan

  were performed. Group 3 had real trailing-whitespace findings corrected before commit. Group 4's broad scan flagged the safe source reference `credential.password`; review confirmed no credential literal.

- Baseline group 5 was then committed as `7ac1229` for the human/audit/manual

  records. No baseline commit was pushed.

Interaction 015 — Phase F review, Git recovery, and Phase G toolchain/pilot preparation

Field	Record
AI tool	Codex CLI
Human instruction received	2026-09-01 17:01:08 +07 (Asia/Ho_Chi_Minh)
Phase G record updated	2026-09-01 17:28:20 +07
Human prompt	Approve Phase F and shared WF-03 invariants; resolve H-002 by human k6-equivalent interpretation; truthfully recover local Git commits; install/pin k6 through existing Homebrew; validate init/output capabilities without SUT HTTP; prepare but do not execute the 2-VU Pilot/helper/runbook. Full actual prompt: `interactions/015-phase-f-review-phase-g-toolchain.md`.
Human-review status	Phase F/H-036 `DONE BY HUMAN`; H-002 `RESOLVED BY HUMAN DECISION`; Pilot H-037 `WAITING FOR HUMAN`
Performance execution	**NONE**
Phase G implementation commit	`d9291f7` — `test: pin k6 and prepare 2-VU pilot`

Actual Git recovery

- Before recovery, Phase 0–F artifacts existed without local procedural commits. No fake/backdated commit, history rewrite, or push was made.
- Five actual baseline commits were created after staged-file, staged-stat, secret/reference, and forbidden-artifact checks: `e488cf4`, `e0d2df5`, `3f1e869`, `1674905`, and `7ac1229`.
- The assignment PDF is protected by an exact root ignore rule. No credential, secret, disposable runtime, or temporary capability file entered Git.
- Branch `main` became five commits ahead of `origin/main`; remotes and `remote.pushDefault=origin` remained unchanged. Nothing was pushed.

Actual installation record

- Preflight found Homebrew 6.0.20 at `/opt/homebrew/bin/brew`, no existing k6, arm64, macOS 26.5.2 build 25F84, and no port-3000 listener.
- Authorized `brew install k6` without sudo installed the 2.2.0 bottle.
- Pinned binary: `/opt/homebrew/bin/k6`; version

`k6 v2.2.0 (commit/devel, go1.26.5, darwin/arm64)`; binary SHA-256 `621ce55919a8067249cfbcc3da4e5ad5316a5706542a9a68667a40a2ec9dbd45`.

- A later `brew info k6` attempted an online formula refresh and failed DNS, but

  still showed installed/linked k6 2.2.0. This failure did not alter the binary.

Genuine no-HTTP compatibility evidence

- A 20-row synthetic credential file and probe scripts existed only under

  `/private/tmp/eshop-hw05-k6-init.a5sx7d`; no value is treated as a real account.

- All official scenario entries passed `k6 inspect --execution-requirements`:

  Load max 5 and 7m30s including graceful stop; Stress exact 13 stages/max 20 and 13m including graceful stop; Spike exact stages/max 20 and 6m35s including graceful stop. Thresholds were absent.

- The prepared Pilot passed after using explicit `-e` variables: 30s to 2,

  three-minute hold, 30s to zero, max 2, 4m30s including graceful stop.

- `k6 deps` resolved local modules plus built-in `k6`, `k6/http`, `k6/data`,

  `k6/execution`, and `k6/metrics`; no custom build was required.

- A no-HTTP runtime probe verified SharedArray/open, one-based VU identity,

  `exec.test.abort`, metric constructors/samples, scenario dispatch, system tags, and summary-hook compatibility. The backend remained stopped.

Genuine output-capability evidence and corrections

- An invalid output-name probe failed before traffic and listed installed output

  types including JSON, CSV, and web dashboard; no file was produced.

- A first one-iteration JSON/CSV probe wrote those private formats, while the

  dashboard bind was sandbox-blocked/too short and no summary file appeared because summary mode was disabled. These failures are retained.

- A corrected two-second no-HTTP probe on an ephemeral local dashboard port

  wrote actual private JSON, CSV, and HTML. A further summary-mode-enabled no-HTTP probe wrote actual private custom-summary JSON.

- These files prove tool syntax/capability only. They are outside Git, are not

  SUT/performance results, and are not homework reports. No JTL was produced; the installed output registry contains no native JTL option.

- The exact 220 KiB probe directory was removed after documentation, including

  its synthetic credential and capability outputs; no repository file was removed.

- The first staged secret-scan command failed before scanning with a genuine

  zsh `bad pattern` quoting error. The corrected simpler scan found only intended source/schema password/token references and no credential literal, private key, common access-token signature, or forbidden staged path.

Phase G draft changes and execution boundary

- Created `docs/k6-toolchain.md`, `runbooks/k6-pilot.md`, internal

  `performance/scenarios/pilot.js`, and unexecuted `performance/tools/provision-accounts.mjs`; updated architecture/output/ resume/manual/human-decision records.

- Pilot samples are isolated as `scenario=pilot,traffic=pilot`; official

  scenarios remain `traffic=measured` and call the same business function.

- The helper uses real registration/login/my-orders contracts, secure random

  passwords, private mode-0600 output outside Git, redacted evidence, and disposable path/marker/database-inode guards. It has not been executed.

- Source revealed that starting this backend resets/seeds its database.

  Therefore the proposed pilot starts the disposable backend once, completes setup-only provisioning, then runs against that same owned process; restarting after setup would erase accounts. H-037 must approve this exact ordering.

No SUT start, port-3000 listener, registration, account, real credential, workflow HTTP, database mutation, Pilot/Load/Stress/Spike/endurance run, measured result, official report, official filename/date, screenshot, JTL, push, or publication occurred.

Interaction 016 — Phase G review and controlled 2-VU Pilot

Field	Actual record
AI tool	Codex CLI
Human approval	One controlled 2-VU disposable Pilot with one-start backend, exactly two accounts, bounded output/evidence, no official interpretation
Actual prompt	Preserved verbatim in [`interactions/016-phase-g-review-2vu-pilot-execution.md`](interactions/016-phase-g-review-2vu-pilot-execution.md)
Artifact ID	`20260901T212619+0700`
Source/tool	Commit `41c6fecf826148e73a4ce3c651791d90650e595c`; k6 v2.2.0 darwin/arm64
Result	**FAILED RUNTIME VALIDATION BEFORE HTTP — NOT OFFICIAL**
Local evidence commit	`11c0a7f` — `test: record failed 2-VU pilot evidence`
Current human gate	H-038 Pilot results/correction review

Preflight and setup evidence

- Attempt 01 preserved a sandbox process-lifecycle failure. The sandboxed backend printed startup lines but did not remain alive; no provisioning or k6 traffic occurred. It was not restarted.

- Attempt 02 used a new clone pinned to `41c6fec`, distinct DB inode, exact

  disposable marker, new evidence paths, >2 GiB free, and locked dependencies.

- The first sandboxed `npm ci` in Attempt 01 failed registry DNS and reported an

  npm exit-handler error. A network-permitted identical lockfile install succeeded. npm reported deprecated `prebuild-install` and four audit vulnerabilities; no automatic fix occurred.

- One externally permitted backend PID 20315 started at 21:30:36 +07, owned

  port 3000, had cwd in Attempt 02, reset/reseeded the clone DB, and remained the same process through setup/Pilot.

- Reset checks: SQLite integrity `ok`, two seed users, five exact products, zero

  orders, valid public rows 01/02, original DB hash unchanged.

- The reviewed helper sequentially provisioned exactly customer keys 01/02 via

  real SUT registration, wrote mode-0600 credentials outside Git, and validated both real logins/role/unlocked/zero-orders. No password/JWT was printed.

Exact Pilot execution and real failure

- The exact four-minute ramping-vus command is preserved with native JSON, CSV,

  summary JSON, dashboard HTML, stdout, and stderr paths.

- k6 reported max 2 VUs, full 4m schedule, 0 interrupted iterations, and

  9,699,772 iterations/attempted/failure samples at 0% workflow success.

- HTTP bytes sent/received were both zero; no login, correlation, check,

  checkout, order, cancellation, or endpoint latency occurred.

- Every iteration failed at the first group boundary. A no-HTTP diagnostic

  confirmed pinned k6's exact cause: `group and check names may not contain '::'`. This is a script/k6 compatibility and runtime/safety failure, not a confirmed SUT bug.

- The generic catch hid the underlying error and allowed an immediate exception

  loop. It generated exact sizes: 13,045,547,159-byte JSON, 5,984,776,809-byte CSV, and 4,161,202,188-byte stderr.

- Custom metrics also used hard-coded `traffic=measured`, while native scenario

  metrics correctly used `traffic=pilot`. `scenario=pilot` still distinguishes them, but the evidence tag is defective.

- The orchestration wrapper closed without returning the numeric Pilot exit code; it is recorded as not captured. A diagnostic demonstrated k6 may return code 0 despite per-iteration script exceptions.

- Scenario execution ended at 4m00s; output completion was about 5m19s from the

  documented command start, roughly 19s beyond the external cap due giant output flushing. This is recorded as an output/runtime safety failure.

Shutdown, evidence, and integrity

- Exact backend PID 20315 received Ctrl-C; harness exit code 1 from interrupt;

  port 3000 then had no listener.

- Disposable DB ended integrity `ok`, four users, two Pilot users, zero orders.
- Original DB SHA-256 remained

  `c63f00544180ba1fbb1427a9b9dd3f1784842698809972f33ce90482e7420ba6` on original device/inode; no original tracked runtime mutation occurred.

- Large files remain local/checksummed and are not staged because they total

  about 21.6 GiB. They were not deleted, renamed, replaced, or called official.

- Small artifacts passed full known-pattern scans. First/last 100 MiB of each

  giant file passed. A complete giant-file scan was stopped after several match-free minutes; complete byte-for-byte secret-scan coverage is not claimed. Zero network data and static metric schemas provide additional bounded evidence.

Correction/review boundary

No source fix and no rerun occurred. H-038 must review: seven safe group-name replacements; custom `context.traffic`; deterministic abort on unexpected runtime/script exception; reliable exit and full wall-clock enforcement; corrected-Pilot output scope; and archival/cleanup of 21.6 GiB local failure evidence. Four deferred numeric abort rules and all final performance thresholds remain deferred. The small evidence/documentation set was truthfully committed locally as `11c0a7f`; credentials and the three giant raw files were excluded. No official scenario or push occurred.

Interaction 017 — Failed-Pilot review and corrected-Pilot authorization

Field	Actual record
AI tool	Codex CLI
Recorded	2026-09-01 22:07:02 +0700
Human review	Failed Pilot confirmed as test-harness/k6 compatibility defect, not SUT performance or a confirmed SUT bug
Full prompt	`interactions/017-failed-pilot-review-corrected-pilot.md`
H-038	**DONE BY HUMAN** after application/commit of the approved corrections
New gate	H-039 corrected-Pilot results review; official execution remains prohibited

Preserved historical transition

- **Old AI/draft:** group names used `wf03::NN_step`, custom metrics hard-coded

  `traffic=measured`, unexpected exceptions were rethrown generically, and no exact-PID runner existed.

- **Observed Pilot failure:** pinned k6 rejected `::`; the generic exception

  path tight-looped 9,699,772 iterations before HTTP, generated about 21.6 GiB, and failed to return a captured numeric process exit.

- **Human-approved correction:** k6-safe group names, context-derived traffic,

  sanitized test-level abort for unexpected harness/runtime exceptions, a five-minute exact-PID runner, bounded Pilot output, and exact deletion of the three untracked bulk files after committed evidence preservation.

Before deletion, the committed artifact inventory preserves all three original paths, exact byte sizes, timestamps, SHA-256 values, producer/output type, exact root error, bounded stderr/diagnostic evidence, failed summary counts, HTTP bytes 0/0, and the no-SUT-request conclusion. The deletion is authorized only after this record and the inventory are committed. One fresh corrected 2-VU Pilot is authorized; no official scenario, threshold, or push is authorized.

Actual correction, cleanup, and blocked corrected attempt

- Static validation found no remaining `::` in executable k6 code. Node/shell

  syntax checks and pinned-k6 init inspection passed with unchanged Pilot stages and tags. A no-HTTP abort probe demonstrated `exec.test.abort()` returns k6 exit code 108 rather than continuing iterations.

- Fix commit `c75b514` contains only the approved names, context traffic tag,

  unexpected-exception abort, exact-PID runner, human/audit records, and pre-deletion evidence. No workflow request/assertion/order/stage changed.

- At 2026-09-01 22:13:05 +07, the three exact authorized untracked files were

  deleted. **FAILED PILOT BULK RAW ARTIFACTS REMOVED AFTER HUMAN-APPROVED EVIDENCE PRESERVATION.** Absence checks passed and filesystem free space increased by 22,674,988 KiB (about 21.62 GiB).

- Corrected Attempt `20260901T221331+0700` used a new clone pinned to `c75b514`,

  new private/output paths, distinct DB inode, clone-only locked dependencies, and one backend PID 22146. PID/cwd/port, reset/seed/product/zero-order/disk, public rows, pinned k6, original hash, and readiness checks passed.

- The first setup-helper invocation was made from the clone and failed safely

  with exit code 1 / `runtime_is_original_repository`. Script-relative `originalRoot` equaled the clone, so the guard stopped before private output, registration, or account creation. Disposable state remained two seed users, zero Pilot users, and zero orders.

- The failure was not silently retried with a different invocation. Owned PID

  22146 was stopped, port 3000 became free, both databases passed integrity, and the original hash remained unchanged. No k6 process, corrected Pilot traffic/result, official scenario, threshold, or push occurred.

Smallest proposal: on a new human-authorized fresh attempt, invoke the same-commit helper from the actual original worktree while passing the clone as `WF03_DISPOSABLE_ROOT`, which matches the helper's protection design. H-039 must review this invocation correction; it is not silently assumed.

The genuine small blocked-attempt evidence and documentation were committed locally as `aebd717` (`test: record corrected Pilot preflight failure`). No credential, disposable DB/runtime, dependency tree, or nonexistent k6 result was included. No push occurred.

Interaction 018 — Invocation-boundary review and fresh corrected Pilot

Field	Actual record
AI tool	Codex CLI
Recorded	2026-09-01 22:35:38 +0700
Human review	Blocked attempt confirmed as provisioning-helper invocation/setup defect; no account/k6 traffic; not SUT performance or confirmed SUT bug
Detailed prompt record	`interactions/018-invocation-boundary-review-fresh-corrected-pilot.md`
Invocation correction	Unchanged helper from actual original worktree; new clone passed explicitly as `WF03_DISPOSABLE_ROOT`
H-039	**DONE BY HUMAN** after this record is committed
Next gate	H-040 fresh corrected-Pilot review; official execution remains prohibited

The human prohibited helper/SUT/workflow/check/correlation/workload/metric changes. Before registration, the original/helper root and new disposable root, different DB inodes, marker, approved commit relationship, original hash, backend PID/cwd/port, and exact localhost target must be recorded. A failed boundary or two-account setup stops the attempt. Only one fresh bounded 2-VU Pilot is authorized after every setup gate passes.

Fresh result

- Documentation-only approval commit `34bb80e` preceded the new clone. The

  original and fresh runtime resolved to different absolute paths and DB inodes, both worktrees were at `34bb80e`, helper hashes matched, the marker was exact, and the original DB hash was unchanged.

- New clone `/private/tmp/eshop-hw05-fresh-corrected-pilot.wyN9jR/runtime`

  started backend PID 23201 once. PID/cwd/port/reset/seed/readiness gates passed.

- The unchanged helper ran from the actual original worktree with the clone as

  `WF03_DISPOSABLE_ROOT`; exit 0. Exactly customers 01/02 were created through real registration and passed private credential, authentication, role, unlocked, and zero-order validation. No helper semantic change occurred.

- Exact-PID runner started k6 PID 23298 and preserved native JSON, summary,

  stdout/stderr, command, metadata, and setup/backend evidence. Numeric k6 exit was 0; runner wall clock 246 seconds; watchdog did not fire; stderr was empty.

- Actual Pilot metrics: 81 attempts/successes; 0 terminal failures; 567 HTTP

  requests with 0 failures; 3,078 checks with 0 failures; 81 orders created and canceled; 0 unexpected-auth samples; lifecycle avg/median/p95 758.33/761/963 ms. All custom samples had `scenario=pilot,traffic=pilot`.

- Runtime checks validated current credential/JWT/identity, exact product match,

  correlated product ID/detail/price, new order ID, same-order pending, cancellation, and final canceled state. Account 01 ended with 36 canceled orders and account 02 with 45, proving both dedicated slots participated.

- Exact backend PID stopped; port free; disposable DB `ok` with all 81 orders

  canceled; original DB `ok` and hash unchanged. Original worktree had no runtime node_modules, credentials, `.env.local`, DB mutation, or temporary secret. Full artifact scan found no password/JWT/Authorization/private email.

These are measured Pilot/runtime-validation observations, not official performance results, thresholds, capacity, or Load conclusions. At that point, H-040 review was required before official-plan finalization. No push occurred.

The genuine fresh Pilot artifact/evidence/documentation set was committed locally as `e23caf4` (`test: record successful corrected 2-VU pilot`). Private credentials, disposable database/dependencies, and secrets were excluded. No push occurred.

Interaction 019 — Successful Pilot approval and Phase I preparation

Field	Actual record
AI tool	Codex CLI
Recorded	2026-09-01 23:03:50 +0700
Human review	Corrected 81/81 Pilot approved as shared-workflow runtime validation only; H-040 `DONE BY HUMAN`
Detailed interaction	[`interactions/019-successful-pilot-review-phase-i.md`](interactions/019-successful-pilot-review-phase-i.md)
Authorized work	Phase I final test-plan review and static official-execution preparation
Prohibited work	No official filenames, Load/Stress/Spike/endurance traffic, final thresholds, screenshots, fake JTL/reports, or push
Current gate	H-010 human creation/approval of the three official attributable filenames

Preserved Pilot correction history

1. The original AI draft used invalid `::` k6 group names. 2. The first Pilot exposed a tight unexpected-exception loop and pathological approximately 21.6-GiB output; it sent no HTTP and was not SUT evidence. 3. Human review corrected naming, context traffic tagging, unexpected-exception test abort, exact-PID wall-clock/exit capture, and bounded Pilot outputs. 4. The next fresh attempt exposed a clone-local provisioning-helper invocation boundary and stopped before account creation or k6. 5. Human review

retained the guard and corrected invocation to the unchanged original-worktree helper targeting a new disposable clone. 6. Fresh corrected Pilot `20260901T223944+0700` passed 81/81 exact WF-03 workflows. The human approved that runtime result without converting it into official performance evidence.

Actual Phase I actions and boundaries

- Promoted only Pilot-exercised shared code/data/helper paths to accurate

  runtime-validated status; official entries remain explicitly unexecuted.
- Reverified the exact Load, Stress, and Spike stage arrays and 30-second

  graceful settings against repository evidence. No stage or threshold changed.
- Created the final historical human-review table, blank official filename

  checklist, deterministic 20-account preconditioning and lockout procedures, three official runbooks, macOS evidence-capture instructions, and genuine text hardware table. No screenshot or filename was generated.
- Prepared one scenario-neutral exact-PID official runner. It accepts only a

  human-created attributable plan, keeps `traffic=measured`, preserves native JSON for every scenario, adds Stress CSV and Spike dashboard, captures numeric exit/watchdog evidence, and contains no business workflow logic.
- Created only trackable scenario result placeholders saying no official result

  exists. No report content, JTL, raw result, credential, disposable runtime, account, backend process, or k6 traffic was generated.
- The first no-HTTP `k6 inspect` loop omitted k6's explicit `-e` flags; the

  existing localhost guard rejected all three entries. The corrected explicit `-e` invocation passed exact configs with `thresholds=null`. The temporary mode-0600 synthetic 20-row file was removed and port 3000 remained free.
- After staged path/secret/disposable/raw/official-filename checks, the truthful

  local Phase I preparation commit was created as `b914c3c` (`test: prepare reviewed official performance plans`). It was not pushed.

Interaction 020 — Human official filenames and blocked wrapper validation

Field	Actual record
AI tool	Codex CLI
Recorded	2026-09-01 23:31:14 +0700
Human action	Created the three official Load/Stress/Spike files under `performance/scenarios/official/` with date `20260901`
Detailed interaction	interactions/020-human-official-filenames-validation.md
Filename outcome	PASS; H-010 `DONE BY HUMAN`

Field	Actual record
Executable-content outcome	BLOCKED on relocated relative paths; no Load preparation followed

All basenames satisfy `^23127027_(Load|Stress|Spike)_20260901\.js$`. SHA-256 comparison showed each official file is byte-for-byte identical to its internal counterpart:

- Load: `6acb45e2f490e0239582f718feafa2ed3e43268cc34a5dab24d9537be9a1dc5b`;
- Stress: `54f5d7329cdecdb45de8c894211019a1c2eeae6250e695bceea809a08ff743a8`;
- Spike: `c3e0049824b7f4b450880fc44c56c2d86e5c0409cbd0d8d3a18701dd77897401`.

The text diff is empty, but the new parent directory changes runtime semantics. `../config/workloads.js` now resolves to the nonexistent `performance/scenarios/config/workloads.js`; equivalent problems affect `../config/runtime.js`, all `../lib/` imports, and `open('../data/workflow.csv')`. Pinned k6 v2.2.0 `inspect` emitted the actual missing-module error for all three wrappers before loading credentials or sending HTTP. Port 3000 remained free.

Codex did not modify, rename, stage, or commit the human-created wrappers. The proposed human-only correction, if retaining the `official/` directory, is to change config/lib/data prefixes from `../` to `../../`. Load workload/output/ account/runtime preparation was deliberately not continued. No official traffic, result, report, screenshot, threshold, account, backend process, or push was created.

Interaction 021 — Human path approval, wrapper validation, and Load preparation

Field	Actual record
AI tool	Codex CLI
Human approval	Exact relative-depth correction approved on 2026-09-01
Continued/application record	2026-09-02 09:05:14 +0700
Detailed interaction	interactions/021-wrapper-correction-load-preparation.md
Wrapper commit	`90fb1ae` (`test: add human-named official k6 wrappers`)
Scope after validation	Official Load preparation only; no execution

Codex applied exactly five string-path changes per official wrapper: `../config/workloads.js`, `../config/runtime.js`, `../lib/data.js`, `../lib/workflow.js`, and `../data/workflow.csv` became their `../../` equivalents. No filename or other line changed.

An ephemeral synthetic 20-row credential file was created mode 0600 under `/private/tmp` only for no-HTTP initialization. Pinned k6 v2.2.0 successfully inspected all three human-named entries. Load showed 0→5/1m, 5/5m, 5→0/1m; Stress and Spike retained their exact approved schedules. Every entry used `traffic=measured`, 30-second graceful settings, the same `runWf03Scenario`, and `thresholds=null`. Normalizing `../../` to `../` produced empty diffs against all internal wrappers. The temporary file was removed and port 3000 remained free.

Only the three official wrappers were staged for the dedicated commit. Staged file and secret scans passed. Commit `90fb1ae` truthfully preserves the human filenames and approved path correction; no push occurred.

Afterward, Load-only preparation pinned the human-named Load entry, exact approved workload/account requirements, one-start fresh-runtime sequence, setup/measured boundary, exact runner invocation, actual-run-ID path contract, native JSON/summary/log/metadata/report/checksum plan, and human Activity Monitor capture instructions. No actual run ID, result artifact, screenshot, backend, account, or traffic was generated. Stress/Spike execution preparation did not proceed.

Interaction 022 — Maximum safe automation authorization

Field	Actual record
AI tool	Codex CLI
Recorded	2026-09-02 09:15:49 +0700
Human instruction	Switch HW05 to maximum safe automation; execute every legitimate automatable task, stop only at explicitly defined human gates, preserve real evidence, never fabricate, and never push upstream
Lockout decision	H-009 human-approved: validated credentials only, no intentional lockout, preserve genuine behavior, no credential substitution, fresh reset/reprovisioning between runs
Execution decision	Human authorized official Load/Stress/Spike/endurance technical execution without further phase approval; real screenshot readiness remains an immediate pre-traffic gate
Git decision	Truthful commits and eventual push to student `origin` authorized after safety/artifact scans
Detailed interaction	`interactions/022-maximum-safe-automation.md`

At receipt, `HEAD` was `70c5429fe82e56e3f48f1c1682f04396e4bc39e1`, `origin` was the student repository, `upstream` was the official SUT, and `remote.pushDefault` was `origin`. No runtime had yet been started. Codex began the authorized official Load preflight and retained the requirement to stop immediately before measured traffic if genuine combined k6/backend-resource visual evidence required human readiness.

The authorization was committed as `e28da7b`. Load preflight then used a fresh clone pinned to that commit. A first sandboxed backend did not persist and sent no account/k6 traffic. Fresh runtime `/private/tmp/eshop-hw05-load.24cMIo/runtime` retained exact backend PID `42059`. One sandbox-networked helper call failed with `sut_request_failed` and created no account/order/credential state; its redacted evidence was preserved. The unchanged helper then succeeded outside the network restriction for exactly 20 validated, unlocked `user` accounts with zero orders. Runtime/database/product/public/private/k6/disk/wrapper/original-integrity gates passed. Actual Load run ID `20260902T092131+0700` was reserved, but its result root, measured traffic, and screenshot do not yet exist. Work stopped only at Human Gate 1 for real macOS screenshot readiness.

Interaction 023 — Official Load execution

Field	Actual record
Human gate input	`READY`
Resume input	`continue` after an interrupted wait
Actual run	Official Load `20260902T092131+0700`
Detailed record	`interactions/023-official-load-execution.md`
Result	Exit 0, watchdog no, 345/345 WF-03 successes

The genuine seven-minute Load produced native JSON/summary/stdout/stderr, command/metadata, redacted setup/backend/postflight evidence, a real human screenshot and real PNG conversion, and a generated aggregate report. Raw-data analysis found 2,415 HTTP requests with zero failures, 13,110/13,110 checks, 345/345 created/canceled orders, 5.731666 requests/s, overall HTTP p95 4.1019 ms and raw p99 4.69974 ms. No threshold or capacity claim was inferred.

After k6 flush, Codex stopped exact backend PID 42059, verified port 3000 free, and verified the protected original DB SHA-256/integrity unchanged. The audit preserves the interrupted wait and corrected post-run process diagnosis rather than rewriting them away.

Interaction 024 — Official Stress preflight

Field	Actual record
Authority	Interaction 022 maximum safe automation
Actual run ID	`20260902T101857+0700`
Detailed record	[`interactions/024-official-stress-preflight.md`](interactions/024-official-stress-preflight.md)
Preflight	PASS; measured traffic not started

Fresh clone `/private/tmp/eshop-hw05-stress.X7FeN0/runtime` is pinned to `0c17457`; exact one-start backend PID `45430` owns the clone listener. Twenty fresh accounts, roles/unlocked/authentication/zero orders, five products, public/private contracts, official wrapper, k6, disk, database boundary, and protected original integrity passed. The new result root remains absent. Work stopped only for the genuine Stress Activity Monitor/k6 screenshot-readiness gate.

Interaction 025 — Official Stress execution and screenshot recovery

Field	Actual record
Human inputs	`READY`; later asked Codex to recheck the screenshot folder
Run	Official Stress `20260902T101857+0700`
Detailed record	[`interactions/025-official-stress-execution.md`](interactions/025-official-stress-execution.md)
Result	Exit 0, watchdog no, 1,281/1,281 workflows

The genuine run produced native JSON and CSV plus summary/log/command/metadata. Analysis measured 8,967 HTTP requests with zero failures, 48,678/48,678 checks, 1,281/1,281 orders created/canceled, p95 3.9147 ms, p99 4.45374 ms, and 11.877557 requests/s. The distinct Stress primary view is a 30-second native- CSV time-series HTML.

The human screenshot was initially in the Load screenshot directory. Codex found and visually verified genuine 20/20-VU Stress/PID `45430` content, moved the unchanged JPEG to the correct run, and generated a real PNG conversion. After flush Codex stopped only PID 45430 and verified port/original integrity. No capacity or threshold was inferred.

Interaction 026 — Official Spike preflight

Fresh Spike runtime `/private/tmp/eshop-hw05-spike.21Nvt8/runtime` is pinned to `eeb02fb`; exact one-start backend PID `48405` owns the clone listener. Twenty fresh accounts, role/unlocked/authentication/zero orders, five products, public/private contracts, official wrapper, k6, disk, DB boundary/integrity, and original hash passed. Actual run ID is `20260902T104549+0700`; result root and traffic remain absent. Detailed record: [`interactions/026-official-spike-preflight.md`](interactions/026-official-spike-preflight.md).

Interaction 027 — Official Spike execution

Field	Actual record
Human inputs	`READY`; later `continue where you stopped`
Run	Official Spike `20260902T104549+0700`
Detailed record	[`interactions/027-official-spike-execution.md`](interactions/027-official-spike-execution.md)
Result	Exit 0, watchdog no, 377/377 workflows

The genuine run produced native JSON/summary, stdout/stderr, exact process metadata, and the real k6 web-dashboard HTML. Analysis measured 2,639 HTTP requests with zero failures, 14,326/14,326 checks, 377/377 orders created and canceled, p95 3.9654 ms, p99 4.91164 ms, and 7.226561 requests/s. Reproducible phase analysis found zero failures during baseline, rise, 20-VU peak, fall, and recovery.

The human-captured screenshot was found in the correct Spike result folder and visually verified as genuine 20/20-VU/PID `48405` evidence. Codex generated a real PNG conversion. After flush it stopped only PID `48405` and verified port and protected-original integrity. No capacity or final threshold was inferred.

Interaction 028 — Evidence-informed endurance design

After all three genuine official scenarios were preserved, Codex selected a conservative five-VU sustained endurance input from actual evidence: official Load had zero failures at five VUs and Stress had zero failures through its bounded 20-VU input. The planned 30s ramp, 12m hold, and 30s ramp-down totals 13 minutes with a 14-minute exact-PID cap. This is an evidence-informed test input, not a final threshold or capacity claim. The entry calls the same shared WF-03 implementation and adds native JSON/CSV output support. Detailed record: [`interactions/028-endurance-design.md`](interactions/028-endurance-design.md). The initial ephemeral no-HTTP fixture used a wrong email pattern and was rejected by the existing guard (k6 inspect exit 107); the audit preserves this AI fixture error rather than presenting the first check as successful.

Interaction 029 — Endurance preflight wrong-listener failure

The fresh endurance attempt produced no k6 traffic. After a sqlite3 native- binding timing failure and a disposable backend launch that did not remain attached, Codex combined listener inspection and provisioning in one shell command without a fail-closed guard. Although the output showed PID `52187` belonged to `/Users/phamngocgiabao/eshop-sut/backend`, provisioning continued and created 20 WF-03 accounts in that unexpected local SUT. Read-only evidence proved the disposable DB still had zero WF-03 accounts and the protected HW05 DB hash was unchanged. Codex did not kill the unowned PID or clean/reset the other DB. Detailed record: [`interactions/029-endurance-preflight-wrong-listener.md`](interactions/029-endurance-preflight-wrong-listener.md).

Interaction 030 — Human stop and blocker resolution

The human asked Codex to check again after requesting PID `52187` be stopped. At 2026-09-02 14:31:38 +0700, read-only checks found the PID absent, port 3000 free, and zero WF-03 accounts in the previously affected external SUT database; its integrity remained `ok`. The protected HW05 DB hash/integrity was unchanged. Codex performed no cleanup or reset. A first malformed AI tool invocation did not execute; the corrected read-only command produced the recorded evidence. H-041 is complete. Detailed record: `interactions/030-endurance-blocker-resolution.md`.

Interaction 031 — Fail-closed disposable listener guard

Codex added a separate exact-PID launcher and verifier after the wrong-listener defect. They require port 3000 to be free, validate that the newly started PID owns it, and require the listener cwd to equal the fresh disposable backend. The existing provisioning helper and all WF-03 semantics remain unchanged. Detailed record: `interactions/031-fail-closed-listener-guard.md`.

Interaction 032 — Fresh endurance preflight

An intermediate fresh runtime proved detached processes are reclaimed by the Codex execution boundary; the exact-PID verifier failed closed and prevented all provisioning/k6 traffic. A second fresh clone, pinned to `4bc3b68`, keeps backend PID `53376` in a foreground session. The atomic PID/cwd/port guard then passed before exactly 20 accounts were provisioned. All account, product, zero-order, DB, k6, workload, disk, source, and original-integrity checks pass. Run ID `20260902T143823+0700` is reserved, but its result root and k6 traffic do not exist. Detailed record: `interactions/032-endurance-fresh-preflight.md`.

Interaction 033 — Endurance execution

The human replied `READY`. Genuine endurance run `20260902T143823+0700` completed 713/713 workflows with zero of 4,991 failed HTTP requests, 27,094/27,094 checks, and 713/713 created orders canceled. k6 exit was 0, watchdog `no`, and wall clock 784 seconds. Raw p95 was 4.377 ms, p99 4.8056 ms, and throughput 6.372423 requests/s. Native-CSV one-minute analysis found no failure or second-half degradation.

The genuine midpoint visual shows 5/5 VUs and exact backend PID `53376` at a point-in-time 1.3% CPU, 56.1 MB, and 11 threads. Codex stopped only that PID after flush and verified port/original integrity. The empirical local endurance point is 5 VUs sustained for 12 minutes; it is not maximum capacity or a final universal threshold. Detailed record: `interactions/033-endurance-execution.md`.

Interaction 034 — Cross-scenario and original AI analysis

Codex reproduced a four-run comparison from committed per-run JSON, totaling 2,716 successful workflows and 19,012 requests with zero failures. It then created the immutable original AI analysis with bounded observations, proposed same-machine regression guardrails, and four source-backed optimization hypotheses. It explicitly avoids capacity, universal-threshold, continuous- resource, or confirmed-bottleneck claims. Human verdicts remain pending and must be recorded separately. Detailed record: `interactions/034-original-ai-analysis.md`.

Interaction 035 — AI analysis and optimization review packet

Codex created factual/raw evidence columns and candidate interpretation concerns for the immutable AI analysis, leaving every final verdict/explanation cell human-owned. It also mapped seven optimization suggestions to actual metrics and source, including explicitly inapplicable/insufficient-evidence possibilities.

No SUT change or synthetic post-optimization result was made. Detailed record: interactions/035-ai-optimization-review-packet.md.

Interaction 036 — Human AI-analysis and optimization verdicts

Phạm Ngọc Gia Bảo directly supplied nine AI-analysis verdicts/explanations and seven optimization verdicts/explanations. Claim 3 is human-classified `MISLEADING`; Claim 8 is `INSUFFICIENT EVIDENCE`. The human also classified a normal index for leading-wildcard search and a generic SQLite connection pool as `HALLUCINATED / NOT APPLICABLE`. The exact human text is preserved in the review matrices and detailed record. `analysis/ai-analysis-original.md` remains unchanged at commit `5e4b00f`. H-014/H-015/H-016 are complete; maximum safe automation resumes. Detailed record: interactions/036-human-ai-optimization-verdicts.md.

Interaction 037 — Technical deliverable automation

Codex created the evidence-based no-confirmed-issue determination, Continuous Performance Testing proposal/Mermaid/undeployed CI prototype, validated Agent Skill, demo preparation, critique evidence/278-word draft, main report, README, PDF renderer/views, and submission validator. The official Skill validator passed after PyYAML was installed only in a temporary `/tmp` virtualenv. No SUT traffic/change, speculative issue, visual, video, URL, grade, final approval, or submission was fabricated. Detailed record: interactions/037-technical-deliverable-automation.md.

Interaction 038 — Submission validation and combined human gate

The real Git log was exported and the validator reported 26 PASS, 0 FAIL, and 9 MANUAL VERIFICATION REQUIRED. Every remaining item is human-attributable: hardware/hostname, video decision/URL, critique approval, no-issue disposition, grade, ZIP approval, and Moodle upload. They were consolidated without invented values in `reviews/FINAL-HUMAN-REVIEW.md`. Detailed record: interactions/038-submission-validation-human-gate.md.

Interaction 039 — Safe origin push

After a clean-worktree, remote, secret, and file-size review, Codex pushed `main` through `0fe6dfc` to the student `origin` only. Push output was `85af3ba..0fe6dfc main -> main`; `upstream` was not modified. Human-only requirements remain pending and no final package/submission was claimed. Detailed record: interactions/039-safe-origin-push.md.

Interaction 040 — Final content approvals and hardware review

The student approved the CPT proposal, Agent Skill, 278-word AI Critique, no-issue disposition, one combined-video format, and hostname compatibility. YouTube URL and grade remain explicitly pending. The supplied genuine hardware JPEG showed correct hostname/hardware but also a device serial; Codex withheld it from Git and requested a human-created safe replacement rather than generatively editing evidence. Human-approved content was promoted and the validator then reported 29 PASS, 0 FAIL, 1 NOT APPLICABLE, and 5 manual checks. Detailed record: interactions/040-final-content-approvals-hardware-review.md.

Interaction 041 — Hardware replacement validation

The student stated that the hardware image had been replaced. Codex inspected the new genuine JPEG at `evidence/hardware/hardware-specs-hostname.jpg`: it is 1778×690 pixels, 100,497 bytes, and has SHA-256 `957b15382a36fe3a1627996d2bdc374ac9a7bfd0c2a1ea9d4f940ea4dc6f6877`. It visibly shows `Phams-MacBook-Pro.local`, MacBook Pro 14-inch, Apple M5, and 16 GB without exposing a serial number or hardware UUID. No AI image editing was performed. H-013 is now `DONE BY HUMAN`; the earlier unsafe candidate and its rejection remain preserved in the historical audit. The validator then reported 30 PASS, 0 FAIL, 4 manual checks, and 1 NOT APPLICABLE; updated report and audit HTML/PDF views were regenerated without changing measurements. After safe staging checks, commit `7d8b5db` was created and the actual push `36abc6b..7d8b5db main -> main` went only to the student `origin`; `upstream` was untouched. Detailed record: `interactions/041-hardware-replacement-validation.md`.